Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS5DS07]

FIFTH SEMESTER B. TECH EXAMINATION, DECEMBER-2021

NOSQL DATABASES

Time: 1:30 Hours **Maximum Marks: 40**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 08 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Enlist the problems encountered in conventional approaches of storing data.	**04**
	b.	Explain Polyglot persistence with suitable example.	**04**
2.	**a.**	Compare and contrast between Open-source software and commercial software.	**04**
	b.	"Scaling up databases by adding a new server or replacing old hardware with new one", discuss.	**04**
3.	**a.**	Define Sharding. Explain different approaches to document record Sharding in short.	**04**
	b.	"ACID consistency is the gold standard of consistency guarantees". Justify the statement.	**04**
		OR	
4.	**a.**	"Pros and cons of NoSQL databases are the roadmap for future developments in database domain", discuss in brief.	**04**
	b.	Write a short note on preventing data loss in NoSQL.	**04**
5.	**a.**	Describe consistent hashing allocation in Voldemort and also explain accessing data partitions.	**04**
	b.	Key-value stores typically act as "buckets" for binary data, State data types in key value store and briefly explain.	**04**
		OR	
6.	**a.**	Storing multiple copies of the same data in other servers, or even racks of servers, helps to ensure availability of data if one server fails. Justify the given statement and also explain the ways in which the replicas can be operated.	**04**
	b.	"Key-value stores typically trade consistency in the data", Explain it in brief.	**04**
7.	**a.**	Illustrate the differences between graph and triple stores.	**04**
	b.	Write a short note on "data provenance".	**04**

OR

8. **a.** "If Leo is the son of Adam, and Adam is the son of Cliff, then you can reason that Leo is the grandson of Cliff. You can "prove" a new fact by referencing others in the database. That is, in order to infer the new triple, you use information available earlier in the text" Explain type of inferencing done in short. **04**

b. "Semantic facts are properly called assertions in triple store". Explain it in brief. **04**